

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 30%

Dr

QUESTIONS			Total Marks: 50
Q.N o	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are	BL1 – Remember	5

	characterized by a unique scheduleNumber, a type (e.g. flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5

4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. ■ A department controls a number of PROJECT. Projects have a unique name, number and a single location. ■ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. ■ Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

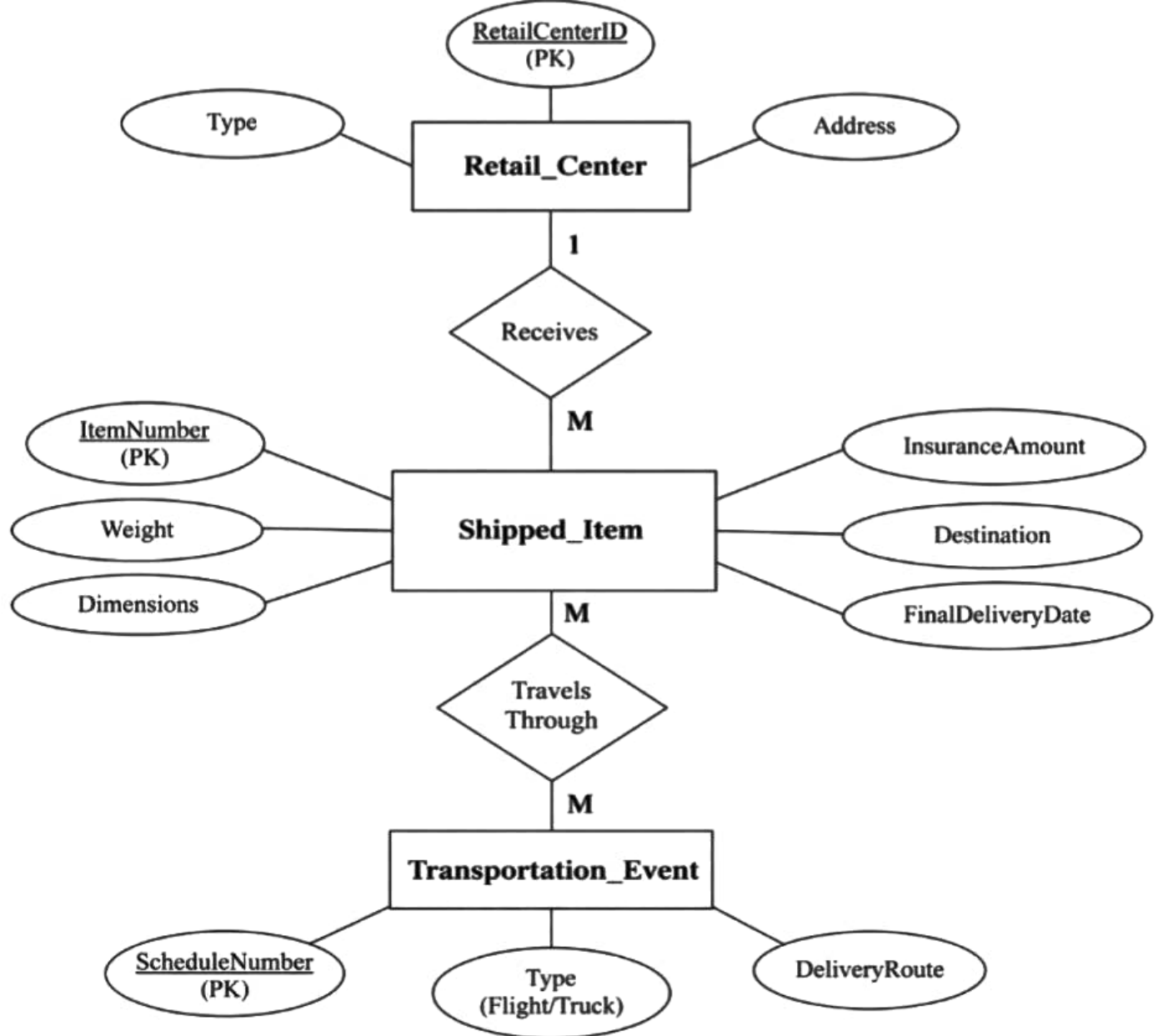
I Niharika(192511221) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

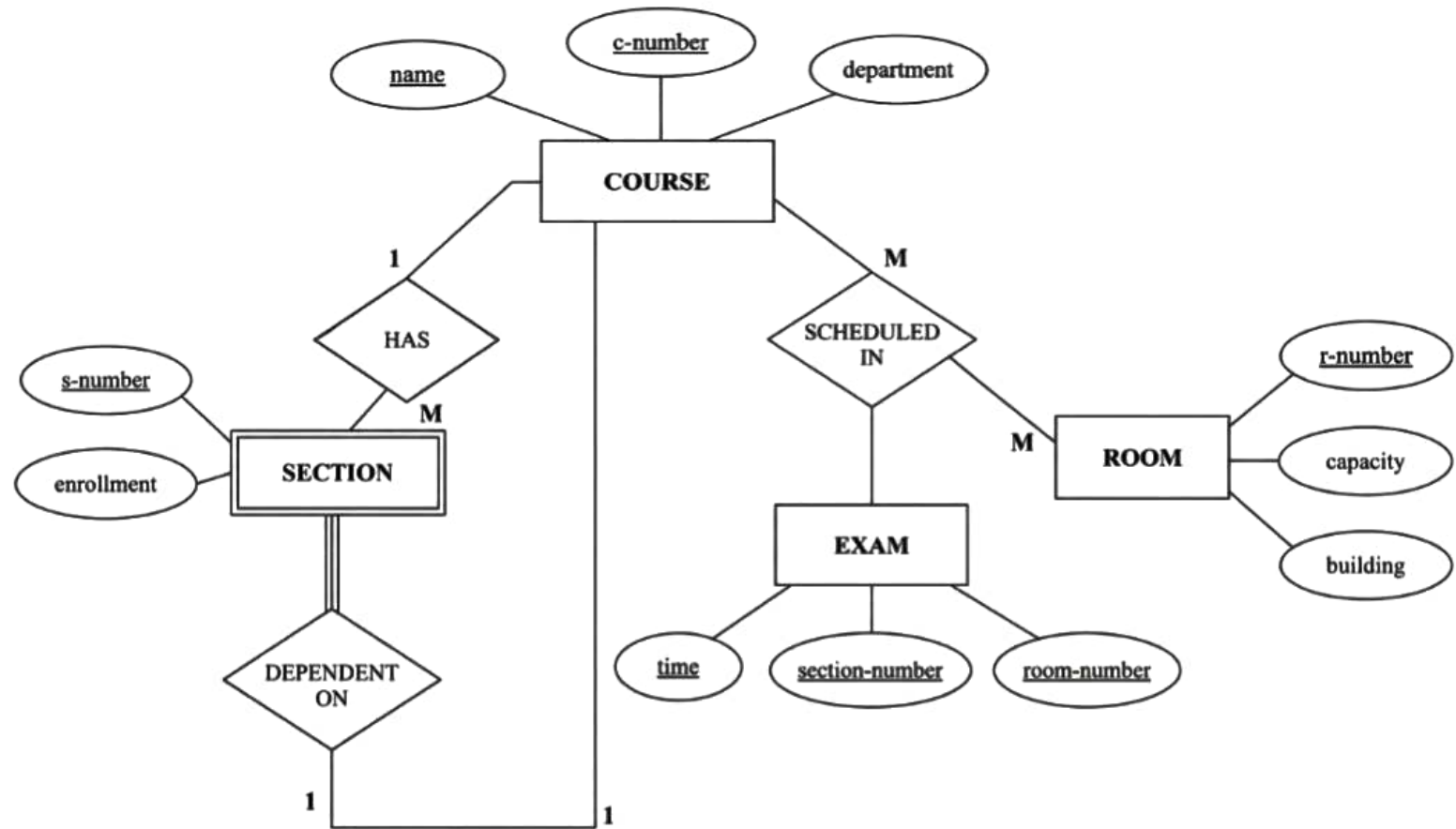
Signature of the student:

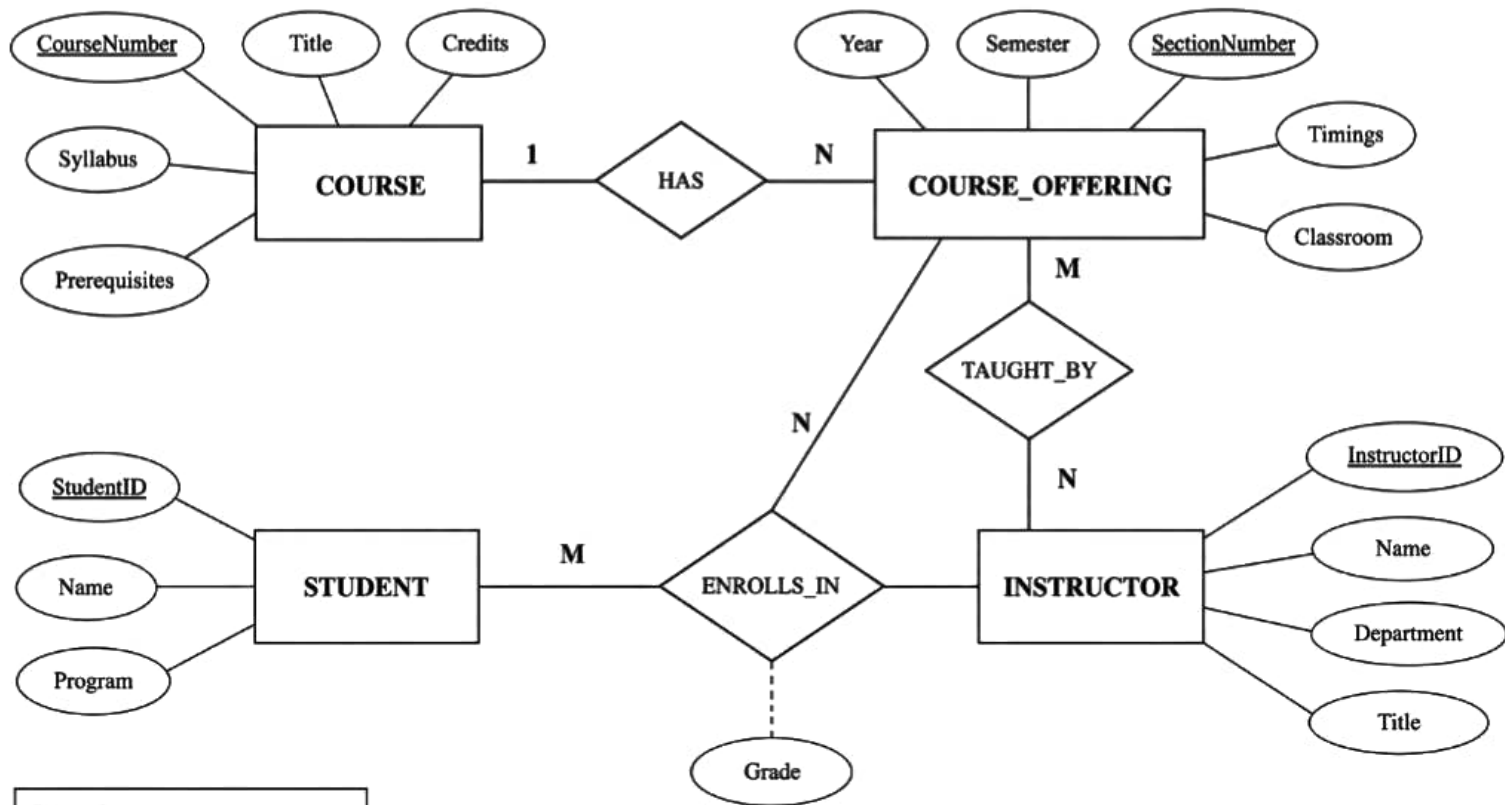
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application

Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand







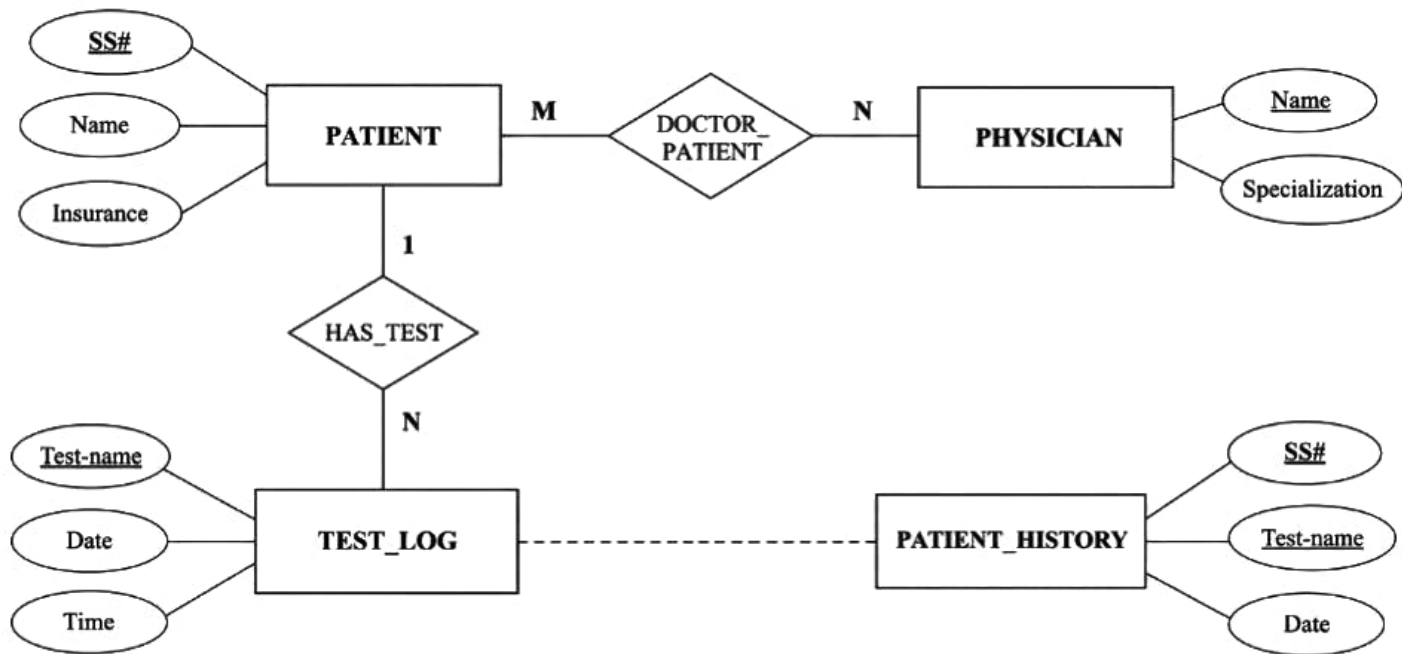
Legend:

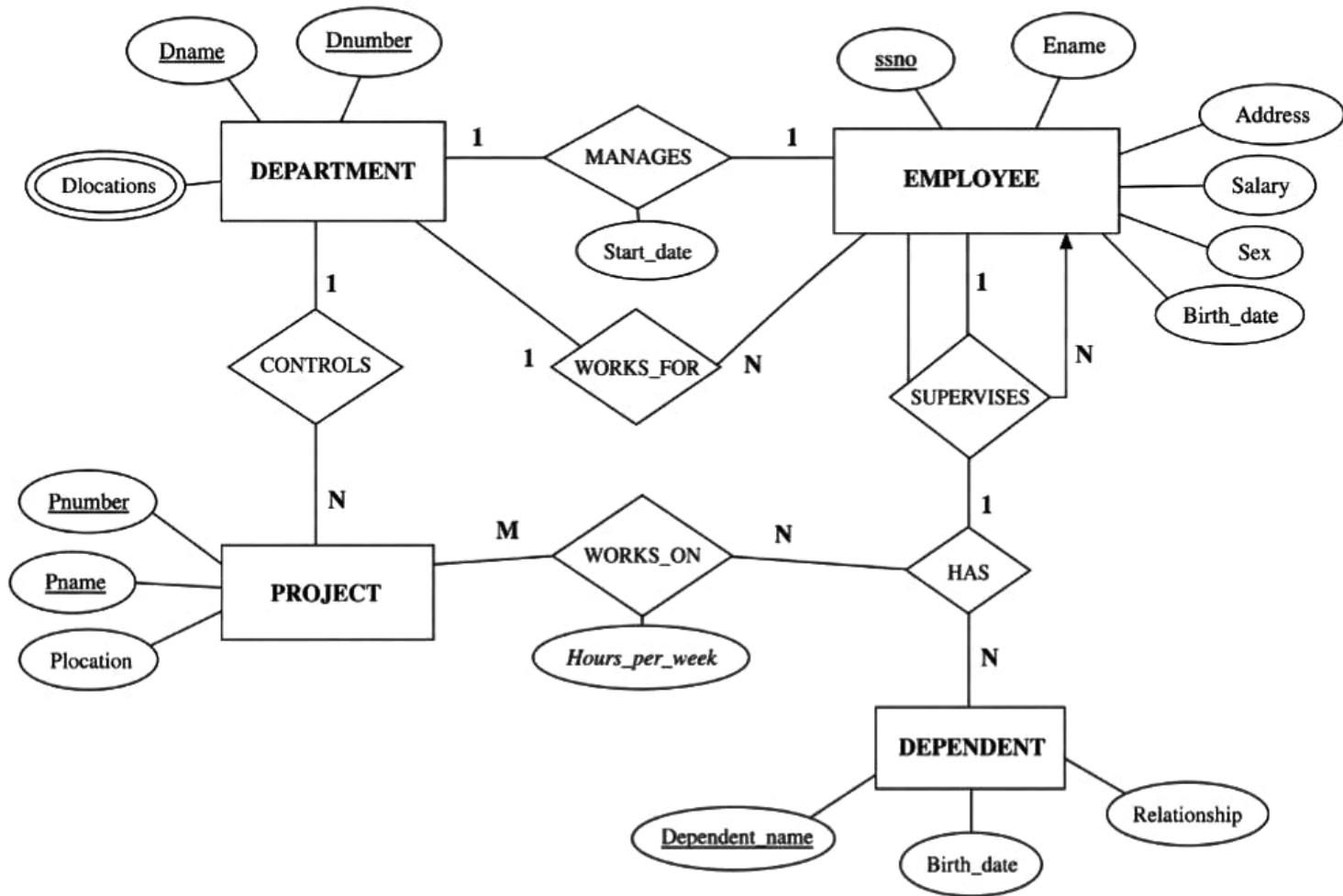
1 : One

N : Many

M : Many (for many)

(Underlined attribute is Primary Key)





1. Explain Schema and Instance with Examples

Schema:

- **A schema is the overall structure or blueprint of a database.**
- **It defines tables, attributes, and relationships.**
- **It changes rarely.**

Example:



Instance:

- **An instance is the actual data stored in the database at a particular time.**
- **It changes frequently as records are added, updated, or deleted.**

Example:

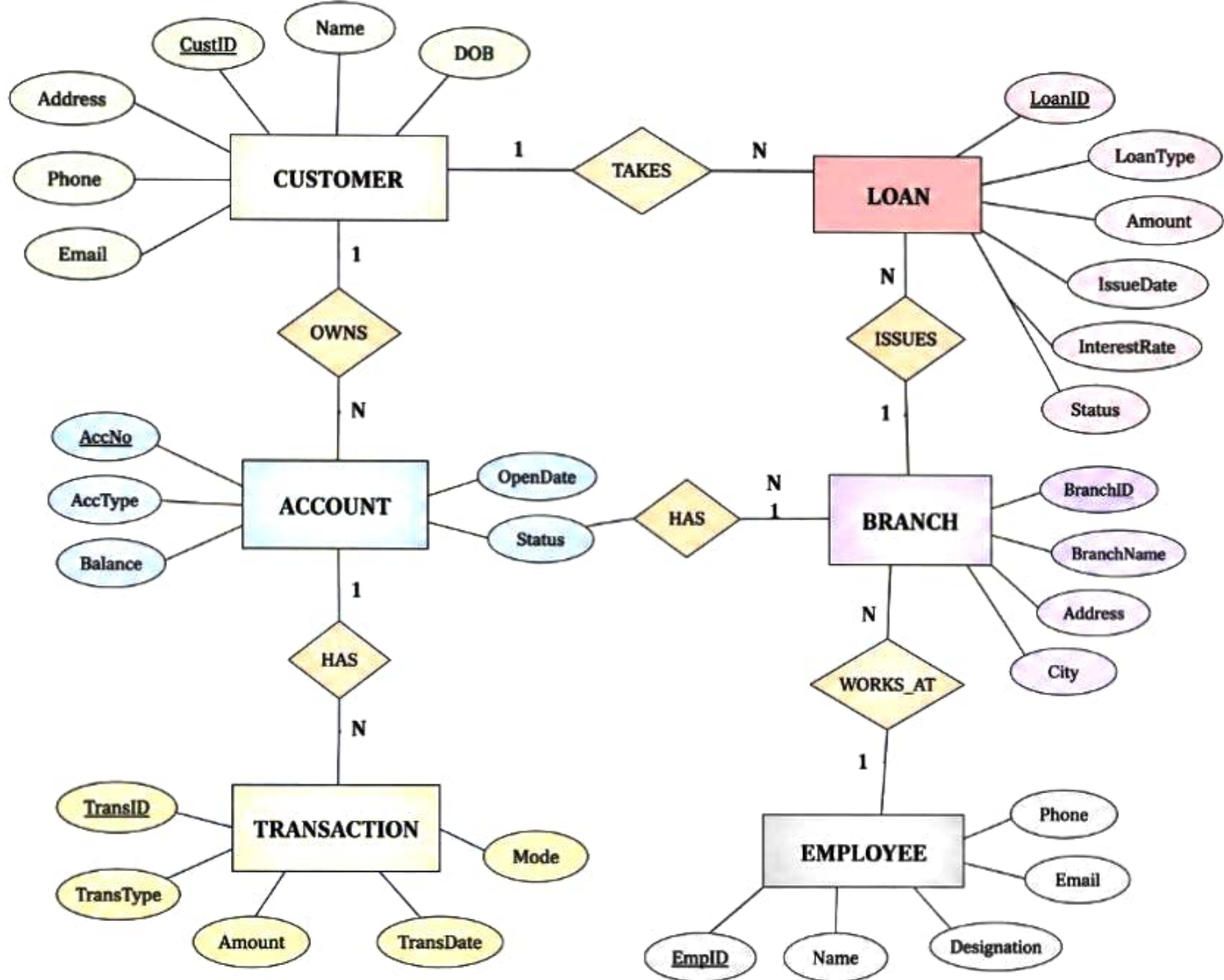
Student_ID	Name	Department
101	Rahul	CSE
102	Priya	ECE

2. Distinguish between Logical Schema, Physical Schema, and View Schema

Logical Schema	Physical Schema
Describes the logical structure of the database.	Describes how data physically stored.
Includes tables, attributes, and relationships.	Includes file organization, indexes, and storage details.
Used by database designers.	Used by database administrators.
Independent of storage details.	Depends on storage techniques.

Example:

- **Logical Schema:** Student(Student_ID, Name, Department)
- **Physical Schema:** Student table stored in a B-tree indexed file on disk.
- **View Schema:** Student(Name, Department) (hides Student_ID).



The Extended Entity-Relationship (EER) Model is an extension of the basic ER model. It provides additional features to represent complex real-world situations more accurately. It includes concepts such as generalization, specialization, and aggregation.

1. Generalization

- **Generalization is the process of combining two or more lower-level entity types into a higher-level entity based on their common attributes.**
- **It follows a bottom-up approach.**

Example:

- **Student and Teacher → Person**

Here, *Person* contains common attributes like Name and Address.

2. Specialization

- **Specialization** is the process of dividing a higher-level entity into lower-level entities based on specific characteristics.
- It follows a **top-down approach**.

Example:

- **Employee → Manager, Engineer, Clerk**

Each specialized entity has its own unique attributes in addition to the common Employee attributes.

3. Aggregation

- **Aggregation** is used when a relationship between entities needs to be treated as a higher-level entity.
- It helps represent relationships among relationships.

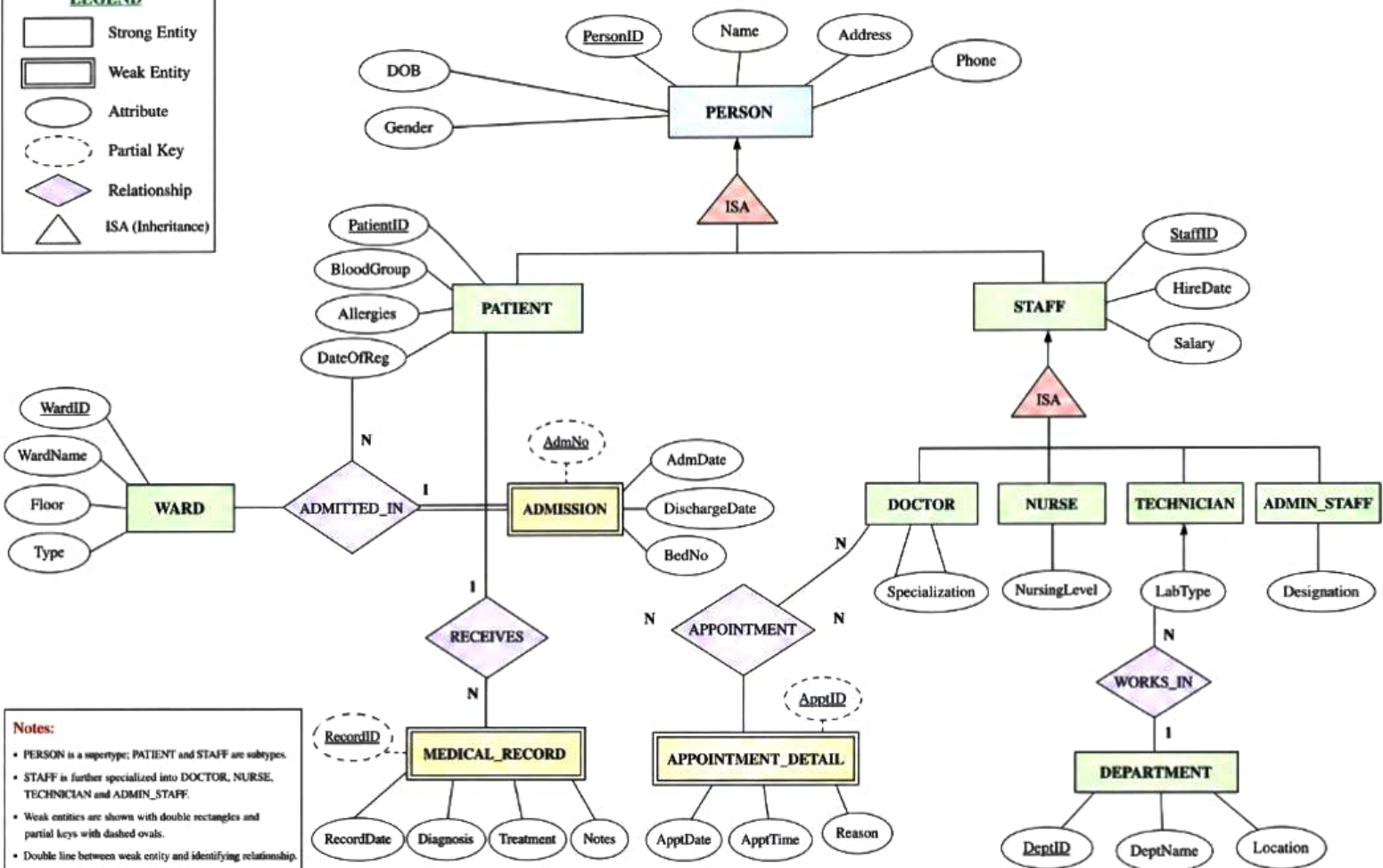
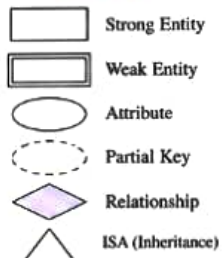
Example:

- **Employee works on Project**
- **Manager monitors the Works_On relationship.**

Here, the **Works_On** relationship is treated as an entity so that another relationship (**Monitors**) can be associated with it.

EER DIAGRAM FOR HOSPITAL MANAGEMENT SYSTEM

LEGEND



Role of a DBA (Database Administrator)

A Database Administrator (DBA) is responsible for managing, maintaining, and securing a database. The DBA ensures that the database is available, performs efficiently, and stores data safely for users and applications.

Responsibilities of a DBA

1. Database Installation and Configuration

- **Installs and configures DBMS software.**

2. Database Design

- **Creates and maintains database schemas, tables, and relationships.**

3. Security Management

- **Controls user access, grants permissions, and protects data from unauthorized access.**

4. Backup and Recovery

- Performs regular backups and restores data in case of system failure.

5. Performance Monitoring

- Optimizes database performance by tuning queries and managing indexes.

6. Data Integrity

- Ensures data is accurate, consistent, and reliable using constraints.

7. Maintenance

- Updates the database, fixes errors, and performs routine maintenance.

8. Storage Management

- Manages disk space and allocates storage efficiently.